

KARAN BHATT

Backend Engineer | Node.js | Microservices | REST APIs

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Professional Summary

Backend-focused Software Engineer with startup experience building scalable applications using Node.js, MongoDB, Redis, Docker, and AWS. Experienced in API design, microservices, asynchronous processing, and system optimization.

Key Achievements

- Built and deployed production applications using Docker, AWS EC2, Redis, and Nginx, with API performance gains from indexing, schema redesign, and Redis caching.
- Solved **350+ DSA problems** on LeetCode with **contest rating of 1692**, demonstrating strong problem-solving across arrays, graphs, DP, and sliding-window patterns.

Work Experience

Adventroz (Avolingo Travel Pvt. Ltd.)

April 2024 – March 2026

Software Engineer (Backend & Product Ownership)

New Delhi, Delhi

- Architected and maintained a microservices-based backend system serving 500+ registered users, with 20+ converting to paid subscriptions.
- Reduced average query response time by ~30–40% through compound indexing and schema redesign across MySQL and MongoDB.
- Led system design and delivery of 3 core product features end-to-end, coordinating across design and QA in 2-week Agile sprints.
- Deployed containerized services using Docker on **AWS EC2** with **Nginx reverse proxy** for routing and scalability.
- Designed API architecture and service communication strategy for scalable system growth.

FitConnectz - Fitness platform connecting people with similar fitness goals

July 2025 – October 2025

Freelance Full-Stack Developer & Architect

Remote

- Designed a scalable backend architecture with **JWT-based authentication** and modular service structure.
- Built a real-time communication system using Socket.io, maintaining sub-200ms message latency for concurrent users.
- Improved API responsiveness through database indexing and REST API optimization.
- Integrated Flutter frontend with backend APIs, ensuring seamless cross-platform performance.
- Handled concurrent user interactions in real-time chat system with consistent low-latency performance.

Key Projects

Lumina AI | *Next.js, TypeScript, Node.js, MongoDB, Redis, BullMQ, WebSockets, Docker, Gemini AI* [Github Link](#)

- Built an AI-powered teaching platform that generates customized exam papers and assignments from teacher-defined parameters using Gemini AI.
- Engineered asynchronous job processing with BullMQ and Redis, eliminating request timeouts on 10–30s LLM generations by moving work to background jobs.
- Implemented JWT authentication with OTP email verification, refresh-token rotation with reuse detection, rate limiting, and HTTP-only cookies.
- Designed an authenticated WebSocket protocol with Origin checks (anti-CSWSH) for real-time progress, and added Gemini resilience via exponential-backoff retries for fault-tolerant generation under load.

Koders Task Platform | *Node.js, WebSockets, Docker, AWS EC2, Redis*

[Github Link](#)

- Containerized both frontend and backend services using Docker, enabling consistent dev-to-production deployment.
- Designed real-time task collaboration using WebSockets with RBAC for secure multi-user access.
- Implemented a Redis caching layer cutting repeated database hits by ~50% on frequently-read endpoints, improving response times.

Technical Skills

Languages: JavaScript (ES6+), TypeScript, Python, Java, C++, Dart

Backend: Node.js, Express.js, REST APIs, WebSockets, JWT Authentication, BullMQ

Frontend: React.js, Next.js, Flutter

Databases & Caching: MongoDB, MySQL, Redis

Testing: Jest, Postman (API testing), manual integration testing

Cloud & DevOps: Docker, AWS EC2, Nginx, Git, GitHub Actions

Core Concepts: System Design, Microservices, Database Optimization, Caching, Queue-Based Architectures

Core CS: DBMS, OS (Basics), Networking (Basics)

Leadership / Extracurricular

IOSC-EDC Club

January 2025 – April 2026

Tech Co-Lead

University School of Automation and Robotics

- Mentored 10+ students in full-stack development, conducting code reviews and technical sessions.
- Led technical execution of 'Vespera', managing a team of 50+ members and improving engagement by 50%.

Education

University School of Automation and Robotics

Sept 2023 – Expected July 2027

Bachelor of Technology (CGPA: 7.84 / 10)

Surajmal Vihar, Delhi